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Smart Thermal System for Patient Safety Monitoring

Physical-Contact (Touch) Detection

Methods Explored and Results

Waveshare 26984 80×62px 3 synchronised views

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June 2026

1. Objective and setup

The touch module must flag *physical contact between patients* from three synchronised thermal cameras (80×62px, 8 Hz), with no optical imagery. Contact is annotated per frame in `contact_labels.csv`: **188 positive frames across only four scenes** (`2ppl_fight`, `2ppl_hug`, `3pp_surprise`, `3pplhedroncolider`); all other scenes are contact-negative, including hard negatives where people are close but not touching (`2ppl_walk`, `3ppl_dance`). All methods are scored on a common per-session timeline split (test = 695 frames, 41 positive). The primary metric is **F1**; recall is the safety priority, false-alarm rate guards against alert fatigue.

2. Approaches tried and results

We evaluated three detector families from the design (multi-view geometric, spatiotemporal graph network, volumetric appearance network), then developed a homography-free pipeline that progressively outperformed them. Table 1 gives the headline F1 on the common test split.

Table 1: Touch detectors on the common test split (single split; see §4 for the cross-validated picture).

Family / method	Idea	F1
Geometric multi-view	boxes $\rightarrow$ homography $\rightarrow$ floor distance	26.8%
MV-STGCN	graph net on projected actor tracks	25.2%
Thermo-X3D	3-D CNN on raw thermal volume (no boxes)	41.7%
<i>Homography-free image-plane pipeline (developed here):</i>		
Box-proximity voting	per-camera box gap + 3-camera vote	28.6%
+ merge-aware / routing	merged-blob counts; people-count routing	35.5%
+ blob-merge	contact = two people in one warm blob	44.6%
+ temporal smoothing	morphological open/close over time	53.8%
+ two-body merge	require exactly-2-body merge in crowds	55.0%
+ raw detector (final)	detector on raw, blobs on residual	60.8%

The final detector (“config-D”) chains: MobileNet-SSD person detection $\rightarrow$ warm-blob segmentation on the background-subtracted residual $\rightarrow$ flag a frame when exactly two people occupy one connected warm component $\rightarrow$ temporal smoothing. It uses **no homography** and needs no contact training labels (thresholds only). It surpasses the deep Thermo-X3D baseline.

3. Key findings

- **Homography is the wrong lever here.** With no surveyed floor points, the homography was self-calibrated from a walking subject; it was accurate enough to localise but too noisy to fuse cleanly, so the geometric and graph methods underperformed (F1 $\approx$ 25–27%).
- **The thermal “merge” signal is decisive.** Two people in contact fuse into one connected warm region; testing for a *two-body* merged blob (rejecting three-body clusters) was the single biggest gain and is simple and explainable.
- **Preprocessing should differ by task.** Background subtraction (Tateno) is *essential* for blob segmentation but *hurts* the person detector — training/running the detector on *raw* frames while segmenting blobs on the *residual* added $\approx$ 6 F1 points.
- **Temporal smoothing helps** (+9 points): contacts persist over time, so removing one-frame false alarms and filling one-frame dropouts is effective.
- **Added complexity did not help.** A learned feature-combiner, motion/optical-flow features, and joint hyper-parameter tuning all matched or *underperformed* the simple rule — consistent symptoms of too few training examples.

4. Honest evaluation: the data limit

A single test split overstates performance. Under 4-fold cross-validation (every frame tested once, no detector leakage) the final detector scores **F1 $\approx$ 44% (fold mean 37%, std $\pm$ 24)**. A bootstrap gives a scene-level 95% confidence interval of **[12%, 66%]** — because positives come from only four scenes, the estimate spans almost the whole range. The 60.8% above is a favourable split, not a reliable figure.

Can more data fix it? We quantified the requirement and tested a shortcut:

- To measure F1 to $\pm 2.5\%$ (so 65% is distinguishable from 60%) needs on the order of $10\times$ **more positive frames and 3–4 $\times$ more distinct contact scenes**.
- **Re-using the older MLX90640 touch data** (real contact, but 24×32 , upsampled) as pretraining *reduced* F1 by 6 points — the low resolution and sensor gap hurt more than the extra events helped. Synthetic compositing was judged unviable (multi-view consistency).

5. Conclusion and recommendation

A practical, explainable, homography-free touch detector exists — MobileNet-SSD + residual blob-merge + temporal smoothing (“config-D”) — which on a matched test split reaches 60.8% F1 and beats the deep baseline. However, the cross-validated estimate ($\approx 44\%$, with a very wide confidence interval) shows the system is **limited by data, not by algorithm**: 188 positive frames in four scenes cannot reliably train richer models or measure performance.

Recommendation: adopt config-D as the working method and prioritise **collecting more contact recordings across more distinct scenarios** (target: several times the current positive count and contact scenes). This is the single change expected to both raise and stabilise performance; further algorithmic tuning on the current data is within measurement noise.